COMSATS University Islamabad, Lahore Campus

Terminal Examination – Fall 2025

Course Title:	Web Technologies			Course Code:	CSC336	Credit	4(3,1)
Course Instructor/s:	Asif Shahzad			Programme Name:	BSCS		
Semester:	6	Batch: FA23	Section:	G1, G2	Date:	Dec. 30, 2025	
Time Allowed:	**3 Hours**			**Maximum Marks:**	**50**		
Student's Name:				Reg. No.			
Note: Please answer only what is asked. Write very neat, keeping the font-size small. No marks for writing stuff that is not asked. You can skip the includes, middle-wares, html, head and body tags etc. Use express and Node.							

Question 1 [2 X 10 = 20 Marks] [CLO-1] Bloom Taxonomy Level: Understanding

a) Write **session usage** related 2 use cases names with 3-4 lines description to explain **why** you think session is needed for them and what information you would store in session. Write **code** of two GET request handlers for URL: /page-1 and /page-2. The /page-1 should **set** a number in session with value 0. The /page-2 should **increment** the number and print updated value on console.
b) Make diagram to explain "**Redirect**" in web applications. Label **diagram** that shall clearly explain related information **flow** between client and server when example1.com redirects to example2.com?

Question 2 [15 Marks] [CLO-2] Bloom Taxonomy Level: Applying

Write HTML and JavaScript for a page with **4 components**: a product category list (Home Gardening, Education, Health and Fitness), a text input to enter search-terms, a button labelled **Search**, and a `div` with id `result`. When the user clicks **Search**, send a **GET** request to `/search` with both input values as query parameters and display the returned HTML inside the `result` div. Write CSS for only label and input tag, they should take 100% width on below 500px screen, and 50% for other screens.

Question 3 [3 X 5 = 15 Marks] [CLO-3] Bloom Taxonomy Level: Applying

a) With ref. to question 2, write handler for **GET** request at `/search` URL, that should **retrieve** values of *category* and *search-terms* from query-string and **use the existing function** `get-products( category, search-terms)` from `lib.js` which returns an array of products having `id, name, price` of product. Render matching **products list** to HTTP client using an **EJS** template.
b) Update the `/search` handler, it should also store the selected product category in a browser **cookie**.
c) Create another handler for `/search-json` URL, which takes same query-string parameters. Make use of `get-products` function but return list of products in **JSON** format to HTTP client.